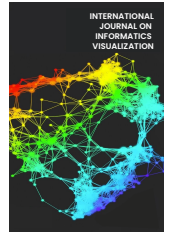




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A Review of Livestock Smart Farming for Sustainable Food Security

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Abstract—Maintaining food security via sustainable farming methods is a significant problem as the global population grows. This study aims to examine the impact of smart farming methods on enhancing farm animal output to satisfy rising demand while fostering sustainability. Smart livestock farming incorporates automation, Internet of Things (IoT) sensors, and machine learning algorithms to improve production, efficiency, and resource utilization. With an emphasis on essential factors including automated feeding, environmental monitoring, and health tracking, this study takes a methodical approach to reviewing IoT-based livestock farming. The efficiency of several sensor technologies, including motion, temperature, humidity, and biometric sensors, is examined in gathering data and making decisions in real time. The potential of machine learning methods like pattern identification, anomaly detection, and predictive analytics to maximize the production and health of farm animals is assessed. According to the results, IoT-driven livestock farming improves illness diagnosis, minimizes resource waste, and optimizes feeding practices, increasing production efficiency. These developments minimize the impact on the environment while promoting steady food production. Additionally, less human interference results from automation in livestock production, which lowers costs and improves decision-making. This study demonstrates how smart agricultural technology may be used to address issues related to food security. Further research is needed to increase real-time data processing, hone machine learning models, and investigate affordable options for broadly adopting these ideas into practice. Livestock management may be transformed, guaranteeing a robust and sustainable agricultural environment.

Keywords—Food security; smart farming; livestock; automation; data analytics; sensor technology.

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I. INTRODUCTION

Amid the increasing global challenges in agricultural and food production management, there is a compelling need to enhance food security on a global scale. Food security emphasizes the significance of concerted efforts among governments, organizations, and communities, utilizing data and technology adoption for automation and effective monitoring in smart farming. This involves optimizing production, distribution, and consumption processes, thereby augmenting adaptability to dynamic circumstances [1], [2]. Research has indicated that the worldwide agricultural sector faces many obstacles, including the expanding global population, which drives up the need for food production and decreases already limited resources. This population rise makes the requirement for additional food to feed billions harder. As a result, the strain on limited resources like crop land, water, and energy is more extensive. The nations have built their own comprehensive National Food Security Framework [3] to handle the difficulties in the agricultural

and food production sectors that aim to safeguard the country's ability to produce, distribute, and receive adequate and healthy food. However, traditional farming practices lack practical monitoring tools, making it challenging to quickly recognize and address livestock's general well-being issues. This affects livestock well-being and decreases production. Furthermore, current farming practices lack data-driven insights, and automation leads to resource utilization and management inefficiency.

Hence, to implement livestock smart farming practices that guide livestock production by promoting an integrated, sustainable, and resilient approach and recognize the interconnection of various components within the food system, the study explores the nuanced dimensions of the pivotal role played by automation, data analytics, and cutting-edge sensor technology in shaping food security strategies. These unique aspects of the practices also need to adopt innovative practices to increase the effectiveness and efficiency of livestock farming globally. This paper reviews academic research papers concerning the global challenges in

livestock, agricultural, and food production management. It discusses suitable cutting-edge technologies and strategies for addressing the trend and gap. The study also proposes a decent framework and guidelines to develop a livestock smart farming that utilizes advanced sensor technologies, data analytics, and automation to monitor livestock health, behavior, and environmental conditions. The proposed framework has the potential to support increasing production on a broader scale across the agricultural sector to optimize resource utilization, decrease environmental impact, and modernize farming practices to contribute to long-term food supply sustainability and adaptability in the face of evolving global demands. The paper is presented as follows: Section I presents the introduction. Section II presents the materials and methods adopted in this study. Section III presents the discussion of results, some future works based on the limitations and gaps identified from the review, and Section IV presents the conclusion.

II. MATERIALS AND METHODS

The technique used for this study is a systematic literature review. The literature review conducted for this study identified the gaps and trends in the use of smart technologies in livestock farming practices, providing a foundation for the development of the livestock smart farming framework. The method requires a thorough and complete review of academic journals on livestock management, sustainability, and the use of smart technology. It examines research papers to determine the most recent developments, practical implementations, and issues involved with using smart technologies in livestock farming practices. It draws significant findings from various information in research journals. This approach not only facilitates the understanding of the existing level of knowledge but also allows for the discovery of gaps, trends, and areas that require further research.

A series of academic research papers and journals from verified indexes and resources about the current systems related to the title [4], [5] was reviewed to understand livestock management and smart technology use in livestock farming for sustainability. The PRISMA guideline by [6] was used for performing and reporting systematic reviews in this investigation, and included the guidelines published by [7] for software engineering and other relevant fields. The subsequent sections provide an overview of the review methodology employed in this paper.

A. Research Question

This study aims to address the challenges that arise in achieving sustainable livestock production and the well-being of farm animals by introducing a sustainable livestock smart farming practice that integrates advanced sensor networks, data analytics, and automation as a solution through a literature-driven assessment. Hence, the study is interpreted with the guidance of 4 research questions (RQ). RQ1: What specific domains are required to monitor livestock farming practices to ensure livestock welfare and increase production for sustainable food security? RQ2: What specific IoT sensor technologies and platforms are suitable for implementing as an optimistic and less manpower-dependent technology in smart livestock farming practices? RQ3: How can data be collected through livestock farming practices to achieve

sustainability and improve livestock welfare for more outstanding production and food security? RQ4: What are the algorithms that have demonstrated effectiveness in analyzing data from livestock farming systems to make predictions to enhance productivity in livestock farming?

B. Digital Sources and Research Strategy

A keyword search string is constructed to accumulate and review relevant research papers, journals, and documents. The keywords “Smart Farming” and the keywords “Food Security” or “Farm Animals” are used as search terms. The entire literature review and research are in verified digital resources and databases, including Scopus, IEEE Xplore, ScienceDirect, SpringerLink, Taylor & Francis, and ProQuest.

C. Selection Criteria

The selection criteria describe a particular method for choosing relevant and appropriate research paper literature. Proper inclusion and exclusion criteria are needed throughout the process to prevent and reduce bias and duplicate literature.

D. Inclusion Criteria

The inclusion criteria for the research study focused on peer-reviewed papers from 2013 to 2023 obtained from web resources. Additionally, only smart livestock (animal) farming systems are included in the study. In the smart livestock (animal) farming system approach, only ruminant animals (cows, sheep, and livestock) are included.

E. Exclusion Criteria

The exclusion criteria for the research study aimed to remove irrelevant articles. Journal papers that lack a comprehensive or significant part of their methodology are not considered for inclusion. Figure. 1 illustrates the PRISMA Flowchart used in this study.

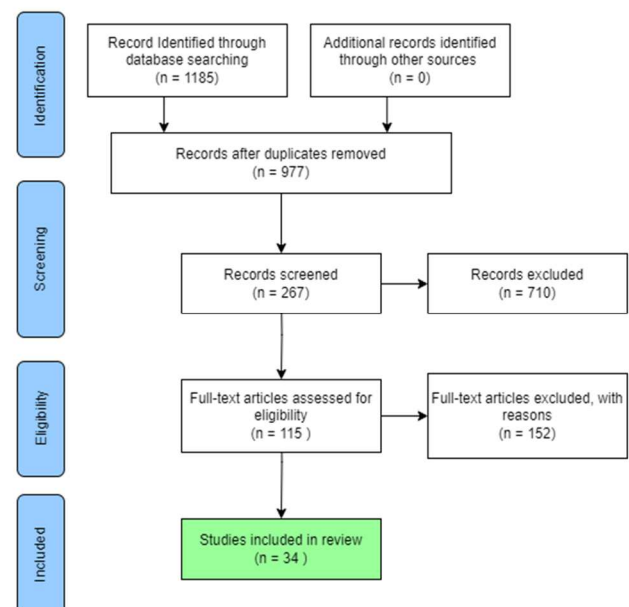


Fig. 1 PRISMA Flowchart

F. Data Extraction Process

Thirty-four papers were selected for the final literature review, and data extraction was conducted on these 34 papers.

III. RESULTS AND DISCUSSION

This section discusses the data from the journals and summarizes the review's findings. Figure 2 displays descriptive data of animal group categorization, revealing that cow, cattle, sheep, and livestock are the most widely studied ruminant animal groups.

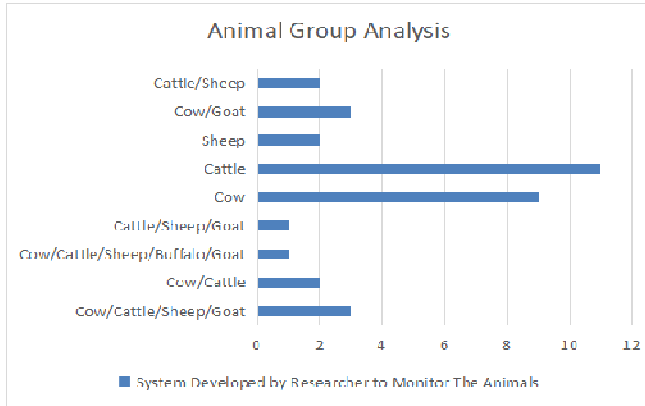


Fig. 2 Ruminant Animal-wise Analysis

Based on the data above, the classification of ruminant animal groups indicates that the most extensively researched ruminant animal groups are cows and cattle, based on the analysis of thirty-four publications. It can be concluded that most research studies were carried out primarily with the cattle and cow category in mind; only a small number of studies were designed to research other ruminant species or to combine different species into their studies. Thus, a greater degree of study approach diversification is needed by including all the ruminant animal groups in the upcoming research to achieve the target of smart farming for sustainable food security.

The following sections offer the results and observations of the systematic literature review based on the research questions.

A. RQ1: What specific domains are required to monitor in livestock farming practices to ensure livestock welfare and increase production for sustainable food security?

RQ1 focuses on understanding the specific domain used in the research of livestock farming practices, with a particular emphasis on protecting livestock welfare for food security. The literature assessment suggests it is essential to investigate strategies for thoroughly monitoring critical elements such as livestock behaviors, health, and environmental conditions. This study attempts to uncover the broad range of tactics and technology used in the quest for sustainable and smart livestock farming by examining research papers.

The data extraction in Table I is based on animal traits, animal behaviors, animal health, and environmental conditions. Animal traits refer to any physical characteristic or feature of a group of animals. Animal behaviors refer to an animal's actions, reactions, or mannerisms in response to environmental stimuli or situations. Animal health refers to the overall state of well-being of an animal, encompassing physical, mental, and social dimensions under a medical approach. Environmental conditions refer to the various factors present in the external surroundings that can influence animals and their ecosystems of natural processes.

TABLE I
DATA EXTRACTION BASED ON RANGE OF DOMAIN

No	Reference	Animal Traits	Animal Behavior	Animal Health	Environmental condition
1	[8]	X	X	X	X
2	[9]				X
3	[10]				X
4	[11]	X	X	X	X
5	[12]	X	X	X	X
6	[13]				X
7	[14]	X	X	X	X
8	[15]		X	X	X
9	[16]	X			
10	[17]			X	X
11	[18]		X		X
12	[19]		X	X	X
13	[20]				
14	[21]	X	X		
15	[22]				X
16	[23]			X	X
17	[24]	X	X	X	X
18	[25]		X	X	X
19	[26]		X	X	X
20	[27]		X	X	X
21	[28]		X	X	
22	[29]		X	X	
23	[30]		X	X	
24	[31]	X	X	X	X
25	[32]	X	X	X	X
26	[33]		X		X
27	[34]	X	X	X	X
28	[35]		X		X
29	[29]		X	X	
30	[36]	X	X	X	
31	[36]		X	X	
32	[37]	X	X	X	X
33	[38]		X		
34	[39]			X	X

The authors have a variety of opinions on their study approach, which can be categorized into animal traits, animal behaviors, animal health, and environmental conditions. Distinctly, almost two-thirds of the researchers conduct their analyses on animal behaviors, animal health, and environmental conditions. The authors [8], [11], [12], [21], [31], [37], [40] emphasize their research study on animal traits. Age of ruminants, growth of the ruminants, weight gain of the ruminants, breeding status of the ruminants, and physical characteristics of the ruminants are some variables the authors explored and discussed in their respective journals. The study of animal behaviors is the most reviewed approach among the parameters by researchers. The authors [8], [11], [12], [15], [19], [21], [25], [26], [28], [31], [33], [35], [37], [40], [41], [42] disputed about animal behaviors like sensation, rumination, movement of animals in the farm, their diet, reactions to insects, pests, and wild animals, rest state, lameness, voluntary and involuntary actions in their study papers.

The research on animal health focuses on diseases like bovine pneumonia and lumpy skin disease recognition, infection identification, heat stress and stroke detection, mental health analysis, and injury discerning [8], [11], [12], [14], [15], [17], [19], [23], [24], [25], [26], [28], [29], [31], [39], [40], [41], [42], [43]. Authors [8], [11], [12], [14], [15], [17], [18], [19], [23], [25], [26], [28], [31], [33], [35], [37], [39], [40], [41], [42], [43] researched the environmental conditions of the animals in their research report. The environmental conditions proposed and analyzed by the

authors include the surrounding area of the farm, the location of the farm, weather conditions, temperature, humidity, air circulation, and many more. Researchers in [11], [12], [14], [24], [31], [37], [40] conducted their research study on all four categories of parameters. Hence, the analysis above shows that mainly the domains of animal behaviors, animal health, and environmental conditions are the favorable scope and field of studies to be investigated to optimize production and sustain food security. The discovery ensures livestock welfare, facilitates optimal production, and contributes to sustainable food security. The response met the research objective of the livestock smart farming practices for achieving sustainable food security. Specific domains such as animal behaviors, animal health, and environmental conditions are required to monitor livestock farming practices to ensure livestock welfare and increase production for sustainable food security. Thus, the research question is answered.

B. RQ2: What specific IoT sensor technologies and platforms are suitable for implementing as an optimistic and less manpower-dependent technology in smart livestock farming practices?

RQ2 emphasizes analyzing research investigating Internet of Things (IoT) sensors and technologies designed to create an optimistic, less human-dependent environment in a smart livestock farming environment. This review highlights the relevance of technology improvements in optimizing livestock farming practices. Existing research articles show a rising priority for integrating IoT technologies to improve productivity, minimize reliance on humans, and provide optimal livestock welfare conditions. This study goes into the vast amount of literature to identify and analyze the wide range of IoT sensors and technologies used or suggested in smart livestock farming, providing insights into their efficacy and applicability within the stated criteria.

TABLE II
DATA EXTRACTION BASED ON MONITORED IOT SENSORS AND TECHNOLOGIES[illegible]

(A) Temperature; (B) Humidity; (C) Moisture; (D) CO2 levels; (E) Odor levels; (F) pH; (G) Air quality; (H) Soil; (I) Weather; (J) Image/Camera; (K) Turbidity; (L) Air speed; (M) Ammonia; (N) Air pressure; (O) Feed level; (P) Water level; (Q) RFID sensor/tag; (R) Biogas; (S) Fire; (T) GPS Device; (U) Heart rate; (V) Respiratory; (W) Motion; (X) Body Pressure; (Y) POF; (Z) PIR; (AA) Raspberry Pi; (AB) Rumen bolus; (AC) UAV; (AD) MQTT; (AE) Node-RED; (AF) Microcontroller; (AG) Accelerometer; (AH) Pedometer; (AI) Blood Pressure; (AJ) Oxygen Level; (AK) Micro Sensor; (AL) Blood glucose; (AM) WiFi; (AN) Pulse Oximeter; (AO) Weight sensor amplifier; (AP) Solar; (AQ) Breath; (AR) Inertial sensor; (AS) SCC sensor

Table II shows the data extraction from 34 selected journals based on the IoT sensors and technologies monitored throughout the articles. A list of IOT sensors and technologies is monitored and displayed in the table with the code below. Based on the journal studies, a vast number of researchers [8], [12], [13], [15], [17], [19], [20], [23], [25], [26], [28], [31], [33], [35], [37], [40], [41], [42] They selected temperature sensors as their IoT sensors to monitor temperature variables.

Temperatures for body and environmental temperatures can be included in both animal behaviors and environmental condition elements.

Meanwhile, humidity sensor [8], [12], [14], [15], [18], [20], [25], [26], [31], [33], [39] image/camera sensor [11], [12], [14], [18], [19], [24], [25], [27], [31], [34], heart rate sensor [12], [17], [19], [20], [23], [26], [27], [31], [37], [39] and accelerometer [12], [14], [19], [20], [24], [25], [27], [28],

[29], [31], [32], [34], [35], [37], [42] are popular IoT sensors among the research. A few researchers picked RFID tags and sensors [8], [12], [14], [26], [27], [32], [39] and GPS technologies [12], [14], [26], [27], [31], [32], [33] to conduct their studies on smart livestock farming. Many authors do not select Air quality, feed, and water monitoring as their observation variables. The air quality of the farm, food, and water levels of animals are observed by [39], who observed the air quality of the farm, while [14] observed the water level using an IoT sensor.

Based on the literature review, a vast number of IoT sensors and technologies can be selected according to the development requirements as parameters in smart livestock farming practices. Still, temperature, humidity, feed consumption, and water intake monitoring sensors are primary sensors that will enhance overall farming activities and manpower requirements. Hence, scholars advocate for the development of resilient smart livestock farming through the integration of temperature, humidity, feed consumption, and water sensors to achieve the objective of the livestock smart farming practices for sustainable food security and thus answer the research question.

TABLE III
LISTS OF PARAMETERS

PARAMETERS	
Temperature	Humidity
Feed	Water

C. RQ3: How can data be collected through livestock smart farming practices to achieve sustainability and improve livestock welfare for more outstanding production and food security?

RQ3 critically investigates how data may be efficiently collected through livestock farming practices to achieve sustainable goals and improve livestock welfare. The literature study provides insight into the different approaches and systems development for the respective IoT devices that collect and utilize data. It investigates the relationship between data gathering techniques and their influence on long-term smart livestock farming practices, offering new insights into innovations that lead to better welfare circumstances. This study attempts to extract findings from current research journals through the viewpoint of RQ3, providing a complete overview of the systems used to gather and run the data collected through IOT components for sustainability and goat welfare development. Table IV reviews systems developed by researchers to enhance livestock farming practices.

TABLE IV
DATA EXTRACTION BASED ON EXISTING SYSTEMS PROPOSED BY RESEARCHERS

Reference	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
[8]	X															
[9]		X	X	X												
[10]					X	X	X	X	X	X						
[11]											X					
[12]												X				
[13]												X				
[14]			X													
[15]													X			
[16]														X		
[17]															X	

Reference	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
[18]																X
[19]													X			
[20]												X				
[21]					X											
[22]	X															
[23]	X															
[24]															X	
[25]	X															
[26]	X															
[27]	X															
[28]															X	
[29]	X															
[30]													X			
[31]	X															
[32]										X						
[33]	X															
[34]												X				
[35]	X															
[43]	X															
[36]															X	
[42]	X															
[37]															X	
[38]	X															
[39]	X															
TOTAL	14	1	2	2	1	1	1	1	1	2	1	4	3	1	5	1

(A) Livestock Monitoring System; (B) Antitheft System; (C) Environment Monitoring System; (D) Feed Monitoring System; (E) Biogas Control System; (F) Feed Control System; (G) Incubator Control System; (H) Fire Detection System; (I) Water Control System; (J) Surveillance System; (K) Support System; (L) No Specific System; (M) Information System; (N) Floor Sensor Pilot System; (O) Health Monitoring System; (P) Irrigation System

A list of systems used to collect data, displayed in the table with code, is provided below the table. The respective researchers solely developed the systems. The general livestock monitoring systems are the most commonly used by researchers [8], [12], [13], [15], [17], [19], [20], [23], [25], [26], [28], [31], [33], [35], [37], [39], [40], [41], [42] to gather and display the data collected from the IoT sensors and technologies. Other than that, specific health monitoring systems developed by researchers [17], [24], [28], [37] efficiently gathered health-based data gained from vast IoT devices. About four researchers [12], [13], [20], [34], there was no specific system to integrate the data, and three researchers [19] developed a basic system that only displays the primary information. The rest of the researchers developed a specific system for their studies according to their respective needs, which compiles very few functions and features. This shows that the livestock monitoring system has gained high demand among the existing systems developed by researchers for the well-being of ruminant animals. Therefore, it would be better and wiser to construct a universal system, such as a livestock monitoring system, soon to support livestock smart farming practices. The adoption of IoT in the proposed system discussed in RQ3 literature should be the benchmark for future research on livestock smart farming practices for sustainable food security and optimization of production.

D. RQ4: What are the algorithms that have demonstrated effectiveness in analyzing data from livestock farming systems to make predictions to enhance productivity in livestock farming?

RQ4 emphasizes the studies on finding specific machine learning algorithms that have proven helpful in analyzing data from smart livestock farming systems, focusing on detecting and avoiding health issues while increasing production. The literature analysis suggests a growing interest in integrating machine learning approaches to maximize the value of data produced from smart livestock farming systems. The literature study, seen from the viewpoint of RQ4, gives a detailed overview of how these algorithms have been used to improve overall production and health outcomes in the field of smart livestock farming, providing significant insights for further research and practical application. The researchers either utilize machine learning algorithms or else no data analysis has been conducted [39]. Table V summarizes whether the researchers did the analysis or used machine learning algorithms to analyze data.

TABLE V
DATA EXTRACTION BASED ON DATA ANALYZED OR NOT

NO OF AUTHORS	NO DATA ANALYSIS	MACHINE LEARNING
[8]	X	
[9]	X	
[10]	X	
[11]	X	
[12]		X
[13]	X	
[14]	X	
[15]		X
[16]		X
[17]		X
[18]		X
[19]	X	
[20]		X
[21]		X
[22]	X	
[23]		X
[24]		X
[25]		X
[26]	X	
[27]		X
[28]	X	
[29]	X	
[30]	X	
[31]		X
[32]	X	
[33]	X	
[34]	X	
[35]		X
[43]		X
[36]	X	
[42]		X
[37]		X
[38]		X
[39]		X
TOTAL	16	18

Based on the table, about sixteen researchers chose not to conduct data analysis, while the remaining eighteen researchers conducted the data analysis using various machine learning algorithms. The deployed machine learning algorithms are a control algorithm [17], support vector machines (SVM) [20], [27], [35], [37], [43], neural network (NN) [20], decision tree[20], [24], [35], [37], long short-term memory [12], [23], [31], deep reinforcement learning and multi-agent deep reinforcement learning [23], multi-layer

perceptron[24], [27], K-nearest neighbor [24], [27], [43], fast Fourier transform[25], logistic regression [37], random forest classifier Naive Bayes, XGBoost, particle swarm optimization, computational intelligence, round robin, equalization technique [39] The 16 authors [8], [12], [13], [15], [17], [19], [20], [23], [25], [26], [28], [31], [33], [35], [37], [40], [41], [42] do not discuss data analysis on the data collection by their respective systems.

This study has proven that a machine learning algorithm is the optimum technique for analyzing the data gathered from the sustainable smart livestock farming system. Researchers stated that specific machine learning algorithms for data analytics were decided based on the data collected from the system of the approach study. Hence, this study confirmed that analyzing and evaluating data collected from the smart livestock farming system to improve livestock well-being and productivity for food security by using a suitable machine learning algorithm is valid.

Based on the research question analysis, this study justified that the fundamental idea behind developing a sustainable framework for smart livestock farming to monitor livestock well-being must consider the combination of innovative and sustainable practices by adopting smart technologies, including sensor technology and data analytics using machine learning. This approach will improve livestock welfare while increasing production, contributing to sustainable food security. Implementing sustainable agricultural practices should emphasize the critical components of integrating smart technology: IoT devices, sensors, and data analytics. The conceptual framework must be designed to monitor and optimize numerous critical variables, including livestock health, behavior, and environmental conditions. It seeks to deliver real-time data, helping farmers to make better decisions and ensure efficient and sustainable livestock farming methods by adopting suitable technology. The conceptual framework for the upcoming livestock smart farming practices and system development includes the concept represented in Figure 3[44], [45].

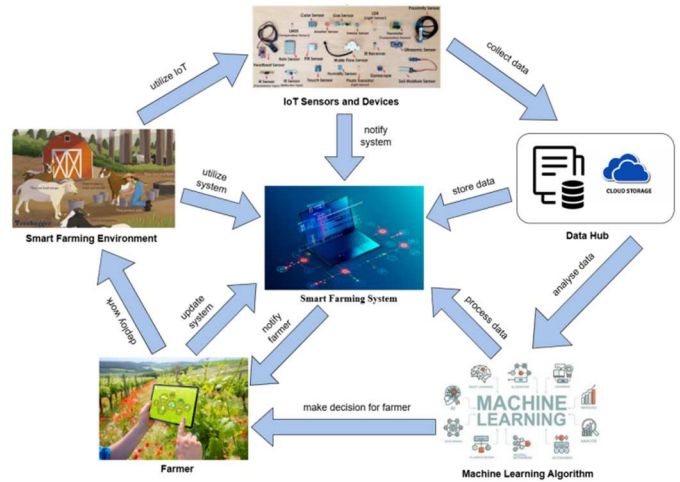


Fig. 3 Conceptual framework of the livestock smart farming ecosystem

IV. CONCLUSION

In conclusion, this study reviews valuable insights and direct implications for food production that significantly influence food security. This paper systematically researched

scholarly works to address questions and objectives to serve as the foundation for research in this area. A systematic literature review that involved a thorough selection process of thirty-four articles was conducted, and the selection was performed based on carefully derived inclusion and exclusion criteria. The findings from this study play a crucial foundation role in the future development and implementation of sustainable smart livestock farming, contributing to the overall goal of ensuring food security.

However, a few challenges arise and are obstacles to achieving the goal. A lack of infrastructure to gather, analyze, and deploy sensors is one of the technological issues facing agriculture today, making customized farming challenging to implement. The absence of skilled workers who are knowledgeable about and capable of implementing cutting-edge technology and policies hinders human capital growth. Workers in agriculture and farming need to be educated and exposed to contemporary methods and equipment. Trade restrictions and political instability make it more difficult to distribute food internationally, which highlights the need for resilient and sustainable agricultural practices. The complications confronting government policy include creating and executing efficient policies, encouraging sustainable practices, and technology. Comprehensive and flexible policies that support innovation while addressing resource constraints and environmental issues are required.

Government agencies, educational institutions, and technology suppliers must work together effectively to overcome these obstacles. This review suggests that sustainable livestock smart farming systems should be thoroughly designed based on the required criteria to meet any country's long-term food supply sustainability and adaptability to a growing population and global demands. The absence of automation complicates process optimization, affecting overall efficiency and sustainability in livestock farming. This highlights the criticality of developing thoroughly designed, sustainable livestock smart farming systems that will address these issues by proactively monitoring livestock activities, promising quick responses according to various parameters based on efficient machine learning data analytics to encourage enhanced animal welfare and production. This initiative is a critical step towards modernizing the industry, maintaining resilience, and establishing progress towards the nation's long-term food security.

Nevertheless, the growing deployment of interconnected IoT devices in urban environments heightens vulnerability to cyber threats. This study also highlights that IoT attacks significantly affect the cybersecurity landscape within the IoT-based implementation of livestock smart farming. In future work, cybersecurity should also be considered to address the cybersecurity issue that could range from safeguarding the genuine IoT node sensors from malicious parties to network security issues in livestock smart farming. The adoption of blockchain technology is proposed as a measure to strengthen the access control mechanism for IoT devices and establish a reliable and robust solution [44], [45], [46], [47], [48].

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REFERENCES

- [1] H. Level Panel of Experts on Food Security, "HLPE Report on Water for food security and nutrition Extract from the Report High Level Panel of Experts on Food Security and Nutrition Extract from the Report I Water for food security and nutrition Water is Central to Food Security and Nutrition (FSN)," 2015. [Online]. Available: www.fao.org/cfs/cfs-hlpe.
- [2] K. Pertanian *et al.*, "Dasar Sekuriti Makanan Negara," 2021.
- [3] KPKM, "Pelan Tindakan Dasar Sekuriti Makanan Negara 2021-2025." [Online]. Available: <https://www.kpk.gov.my/pelan-tindakan-dasar-sekuriti-makanan-2021-2025>
- [4] Institute for Academic Development, "What is a literature review?" *The University of Edinburgh*, [Online]. Available: <https://institute-academic-development.ed.ac.uk/study-hub/learning-resources/literature-review>. [Accessed: Sep. 21, 2024].
- [5] "Research Guides: Systematic reviews: home." [Online]. Available: <https://guides.library.ucla.edu/systematicreviews>
- [6] D. Moher, A. Liberati, J. Tetzlaff, and D. G. Altman, "Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement," *BMJ*, vol. 339, no. jul21 1, pp. b2535–b2535, Jul. 2009, doi: 10.1136/bmj.b2535.
- [7] B. Kitchenham *et al.*, "Systematic literature reviews in software engineering – A tertiary study," *Inf Softw Technol*, vol. 52, no. 8, pp. 792–805, Aug. 2010, doi: 10.1016/j.infsof.2010.03.006.
- [8] L. Yang, X. Y. Liu, and J. S. Kim, "Cloud-based Livestock Monitoring System Using RFID and Blockchain Technology," in *Proceedings - 2020 7th IEEE International Conference on Cyber Security and Cloud Computing and 2020 6th IEEE International Conference on Edge Computing and Scalable Cloud, CSCloud-EdgeCom 2020*, Institute of Electrical and Electronics Engineers Inc., Aug. 2020, pp. 240–245. doi:10.1109/CSCloud-EdgeCom49738.2020.00049.
- [9] A. R. Arko, S. H. Khan, and M. S. Islam, "Smart Security, Food and Environment Monitoring System for Cows using Microcontroller," 2019 International Conference on Communication and Electronics Systems (ICCES), pp. 679–684, Jul. 2019, doi:10.1109/icc45898.2019.9002293.
- [10] A. Easwaran, P. Arvindan, E. Dhanyasree, R. Surya, and S. Selvakumar, "Internet of Things Enabled Smart Animal Farm Prototype," *Journal of Physics: Conference Series*, vol. 2070, no. 1, p. 012115, Nov. 2021, doi: 10.1088/1742-6596/2070/1/012115.
- [11] K. Noinan, S. Wicha, P. Surepong, and R. Chairicharoen, "Requirements Analysis and Design for Thai Beef-Cattle Farm Management System," in *2021 Joint 6th International Conference on Digital Arts, Media and Technology with 4th ECTI Northern Section Conference on Electrical, Electronics, Computer and Telecommunication Engineering, ECTI DAMT and NCON 2021*, Institute of Electrical and Electronics Engineers Inc., Mar. 2021, pp. 356–359. doi: 10.1109/ectidamtcon51128.2021.9425779.
- [12] N. Valov, T. Mladenova, and I. Valova, "IoT and big data in animal farming," in *ISMSIT 2021 - 5th International Symposium on Multidisciplinary Studies and Innovative Technologies, Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2021, pp. 93–96. doi: 10.1109/ismsit52890.2021.9604671.
- [13] M. Billah *et al.*, "Real-time Monitoring of Water Quality in Animal Farm: An IoT Application," 2019, pp. 27–29. doi: 10.1109/icsima47653.2019.9057320.
- [14] M. Alexy and T. Haidegger, "Precision Solutions in Livestock Farming - Feasibility and applicability of digital data collection," in *ICCC 2022 - IEEE 10th Jubilee International Conference on Computational Cybernetics and Cyber-Medical Systems, Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 233–238. doi: 10.1109/ICCC202255925.2022.9922883.
- [15] I. Valova and T. Mladenova, "An Information System for Livestock and Pasture Surveillance," in *13th National Conference with International Participation, ELECTRONICA 2022 - Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2022. doi:10.1109/electronica55578.2022.9874425.
- [16] J. Vaughan, P. M. Green, M. Salter, B. Grieve, and K. B. Ozanyan, "Floor sensors of animal weight and gait for precision livestock

- farming,” 2017 IEEE Sensors, pp. 1–3, Oct. 2017, doi:10.1109/icsens.2017.8234202.
- [17] Y. Huang, C. Rao, K. Cao, J. Nie, J. Hu, and B. Tan, “Health Monitoring System for Dairy Farming,” in *2023 IEEE 3rd International Conference on Electronic Communications, Internet of Things and Big Data, ICEIB 2023*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 82–86. doi:10.1109/ICEIB57887.2023.10170269.
- [18] N. K. Bhanu and K. Sahana, “Farm Vigilance: Smart IoT System for Farmland Monitoring and Animal Intrusion Detection using Neural Network,” in *2021 Asian Conference on Innovation in Technology, ASIANCON 2021*, Institute of Electrical and Electronics Engineers Inc., Aug. 2021. doi: 10.1109/asiancon51346.2021.9544926.
- [19] E. Paunova-Hubenova and E. Trichkova-Kashamova, “Smart Solutions for Control and Management in Livestock Farms,” in *Big Data, Knowledge and Control Systems Engineering - Proceedings of the 7th International Conference, BdKCSE 2021*, Institute of Electrical and Electronics Engineers Inc., 2021. doi:10.1109/bdkcse53180.2021.9627269.
- [20] R. F. Aunindita, M. Shiam Misbah, S. Bin Joy, M. A. Rahman, S. I. Mahabub, and J. Noor Mukta, “Use of Machine Learning and IoT for Monitoring and Tracking of Livestock,” in *Proceedings of 2022 25th International Conference on Computer and Information Technology, ICCIT 2022*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 815–820. doi: 10.1109/ICCIT57492.2022.10055766.
- [21] R. Sokullu, B. Y. Tanriverdi, and R. Goleva, “IoT Based Livestock Precision Feeding System Using Machine Learning,” in *2022 8th International Conference on Energy Efficiency and Agricultural Engineering, EE and AE 2022 - Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2022. doi:10.1109/EEAE53789.2022.9831244.
- [22] M. Bovo et al., “A Smart Monitoring System for a Future Smarter Dairy Farming,” 2020 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor), pp. 165–169, Nov. 2020, doi: 10.1109/metroagrifor50201.2020.9277547.
- [23] Q. Zhang, Y. Mahajan, I. R. Chen, D. S. Ha, and J. H. Cho, “An Attack-Resilient and Energy-Adaptive Monitoring System for Smart Farms,” in *Proceedings - IEEE Global Communications Conference, GLOBECOM*, 2022, pp. 2776–2781. doi:10.1109/globecom48099.2022.10001060.
- [24] D. Kaur and A. Kaur, “IoT and Machine Learning-Based Systems for Predicting Cattle Health Status for Precision Livestock Farming,” in *2022 International Conference on Smart Generation Computing, Communication and Networking, SMART GENCON 2022*, Institute of Electrical and Electronics Engineers Inc., 2022. doi:10.1109/smartgencon56628.2022.10083995.
- [25] M. Caria et al., “Smart Farm Computing Systems for Animal Welfare Monitoring,” 2017. doi: 10.23919/MIPRO.2017.7973408.
- [26] J. A. Anand, K. Pawan, M. Narendran, M. Johnson, and S. Rakesh, “Real Time Livestock Rearing,” in *2022 1st International Conference on Computer, Power and Communications, ICCPC 2022 - Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 603–606. doi: 10.1109/iccpc55978.2022.10072153.
- [27] A. Yaseer and H. Chen, “A Review of Sensors and Machine Learning in Animal Farming,” in *2021 IEEE 11th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems, CYBER 2021*, Institute of Electrical and Electronics Engineers Inc., Jul. 2021, pp. 747–752. doi:10.1109/cyber53097.2021.9588295.
- [28] S. Debdas, A. Behera, A. Bandyopadhyay, S. Karmakar, and A. Subhadarshini, “An IOT Solution for Cattle Health Monitoring and Tracking,” in *Proceedings - 2022 OITS International Conference on Information Technology, OCIT 2022*, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 513–518. doi:10.1109/OCIT56763.2022.00101.
- [29] M. Taneja, J. Byabazaire, A. Davy, and C. Olariu, “Fog Assisted Application Support for Animal Behaviour Analysis and Health Monitoring in Dairy Farming,” 2018. doi: 10.1109/WF-IoT.2018.8355141.
- [30] M. H. Ariff, I. Ismail, “Livestock Information System using Android Smartphone,” in *Livestock Information System using Android Smartphone*, IEEE, 2013, p. 322. doi: 10.1109/SPC.2013.6735123.
- [31] K. Dineva, D. Parvanov, T. Atanasova, G. Mateeva, P. Petrov, and G. Kostadinov, “Towards CPS/IoT System for Livestock Smart Farm Monitoring,” in *International Conference Automatics and Informatics, ICAI 2021 Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2021, pp. 252–255. doi:10.1109/ICAIS2893.2021.9639460.
- [32] N. Markov et al., “Smart Dairy Farm - Digitalization and Innovation,” in *2022 8th International Conference on Energy Efficiency and Agricultural Engineering, EE and AE 2022 - Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2022. doi:10.1109/EEAE53789.2022.9831220.
- [33] C. Joshitha, P. Kanakaraja, M. D. Bhavani, Y. N. V. Raman, and T. Sravani, “Lorawan based cattle monitoring smart system,” in *Proceedings of the 7th International Conference on Electrical Energy Systems, ICEES 2021*, Institute of Electrical and Electronics Engineers Inc., Feb. 2021, pp. 548–552. doi:10.1109/ICEES51510.2021.9383749.
- [34] L. Angrisani et al., “BUFF4L.0: Veterinary and Engineering Sciences at the crossroads in the Industry 4.0 age,” in *6th International Forum on Research and Technology for Society and Industry, RTSI 2021 - Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2021, pp. 137–141. doi: 10.1109/RTSI50628.2021.9597308.
- [35] T. H. Dang, N. H. Dang, V. T. Tran, and W. Y. Chung, “A LoRaWAN-Based Smart Sensor Tag for Cow Behavior Monitoring,” in *Proceedings of IEEE Sensors*, Institute of Electrical and Electronics Engineers Inc., 2022. doi: 10.1109/sensors52175.2022.9967209.
- [36] M. H. Ariff, I. Ismarani, N. Shamsuddin, “RFID based systematic livestock health management system,” IEEE, 2014, p. 189. doi:10.1109/SPC.2014.7086240.
- [37] A. Trivedi and P. S. Chatterjee, “CARE: IoT enabled Cow Health Monitoring System,” in *2022 2nd International Conference on Intelligent Technologies, CONIT 2022*, Institute of Electrical and Electronics Engineers Inc., 2022. doi:10.1109/conit55038.2022.9847701.
- [38] Luis Nóbrega; André Tavares; António Cardoso; Pedro Gonçalves, “Animal monitoring based on IoT technologies,” IEEE, 2018. doi:10.1109/iot-tuscany.2018.8373045.
- [39] A. Ur Rehman et al., “Implementation of an Intelligent Animal Monitoring System Using Wireless Sensor Network and IoT Platform,” in *International Conference on Cyber Resilience, ICCR 2022*, Institute of Electrical and Electronics Engineers Inc., 2022. doi:10.1109/ICCR56254.2022.9996080.
- [40] L. Angrisani et al., “BUFF4L.0: Veterinary and Engineering Sciences at the crossroads in the Industry 4.0 age,” in *6th International Forum on Research and Technology for Society and Industry, RTSI 2021 - Proceedings*, Institute of Electrical and Electronics Engineers Inc., 2021, pp. 137–141. doi: 10.1109/RTSI50628.2021.9597308.
- [41] A. Yaseer and H. Chen, “A Review of Sensors and Machine Learning in Animal Farming,” in *2021 IEEE 11th Annual International Conference on CYBER Technology in Automation, Control, and Intelligent Systems, CYBER 2021*, Institute of Electrical and Electronics Engineers Inc., Jul. 2021, pp. 747–752. doi:10.1109/CYBER53097.2021.9588295.
- [42] A. A. Chaudhry, R. Mumtaz, S. M. Hassan Zaidi, M. A. Tahir, and S. H. Muzammil School, “Internet of Things (IoT) and Machine Learning (ML) enabled Livestock Monitoring,” in *HONET 2020 - IEEE 17th International Conference on Smart Communities: Improving Quality of Life using ICT, IoT and AI*, Institute of Electrical and Electronics Engineers Inc., Dec. 2020, pp. 151–155. doi:10.1109/HONET50430.2020.9322666.
- [43] Z. Shi et al., “Internet-of- Things Behavior Monitoring System Based on Wearable Inertial Sensors for Classifying Dairy Cattle Health Using Machine Learning,” in *5th IEEE International Conference on Artificial Intelligence in Engineering and Technology, IICAIET 2023*, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 277–282. doi: 10.1109/IICAIET59451.2023.10291766.
- [44] M. Noorafiza, H. Maeda, R. Uda, T. Kinoshita, and M. Shiratori, “Vulnerability analysis using network timestamps in full virtualization virtual machine,” in *ICISSP 2015 - 1st International Conference on Information Systems Security and Privacy, Proceedings*, SciTePress, 2015, pp. 83–89. doi: 10.5220/0005242000830089.
- [45] N. A. M. Razali et al., “Political Security Threat Prediction Framework Using Hybrid Lexicon-Based Approach and Machine Learning Technique,” *IEEE Access*, vol. 11, pp. 17151–17164, 2023, doi: 10.1109/access.2023.3246162.
- [46] W. N. W. Muhamad et al., “Evaluation of Blockchain-based Data Sharing Acceptance among Intelligence Community,” *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 12, pp. 597–606, 2020, doi: 10.14569/IJACSA.2020.0111270.
- [47] N. M. Noor et al., “Decentralized Access Control using Blockchain Technology for Application in Smart Farming,” *IJACSA*

International Journal of Advanced Computer Science and Applications, vol. 13, no. 9, p. 2022, 2022, doi:10.14569/ijacsa.2022.0130993.

[48] W. Nurhidayat, W. Muhamad, N. Matrazali, K. Khalil Ishak, and S. Ramli, "Secure Blockchain-Based Data-Sharing Model and Adoption among Intelligence Communities," *Article in IAENG International Journal of Computer Science*, 2021.